

HDOC DAY-13

Menu-Driven C Program Using Switch-Case

Name	Bhavya Rana
SAP ID	590034689
Batch	15

1. Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Count the number of even digits and odd digits.
2. Find the sum of even digits and the sum of odd digits separately.

Examples: 12045 -> Even_dig = 3, Odd_dig = 2 12345 -> Sum_even = 6, Sum_odd = 9

2. Algorithm

1. Start and input a positive integer n.
2. Display the menu for the two operations.
3. Read the user's choice and use switch(choice).
4. For choice 1, initialize even_dig = 0, odd_dig = 0 and temp = n.
5. While temp > 0, get digit = temp % 10. Increment even_dig if even; otherwise increment odd_dig. Set temp = temp / 10.
6. Display Even_dig and Odd_dig.
7. For choice 2, initialize sum_even = 0, sum_odd = 0 and temp = n.
8. While temp > 0, get digit = temp % 10. Add even digits to sum_even and odd digits to sum_odd. Set temp = temp / 10.
9. Display Sum_even and Sum_odd.
10. For another choice, display Invalid choice.
11. Stop.

3. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	12045 Choice 1	Even_dig = 3 Odd_dig = 2	Even_dig = 3 Odd_dig = 2	Pass
2	12345 Choice 2	Sum_even = 6 Sum_odd = 9	Sum_even = 6 Sum_odd = 9	Pass

4. C Program - Code

```
UW PICO 5.09 File: day13.cpp

#include <stdio.h>

int main()
{
    int n, choice, digit;
    int even_dig, odd_dig;
    int sum_even, sum_odd;
    int temp;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    printf("\nMENU\n");
    printf("1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits separately\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            even_dig = 0;
            odd_dig = 0;
            temp = n;

            while (temp > 0)
            {
                digit = temp % 10;

                if (digit % 2 == 0)
                    even_dig++;
                else
                    odd_dig++;

                temp = temp / 10;
            }

            printf("Even_dig = %d\n", even_dig);
            printf("Odd_dig = %d\n", odd_dig);
            break;

        case 2:
            sum_even = 0;
            sum_odd = 0;
            temp = n;

            while (temp > 0)
            {
                digit = temp % 10;

                if (digit % 2 == 0)
                    sum_even = sum_even + digit;
                else
                    sum_odd = sum_odd + digit;
            }
    }
}
```

⌘ Get Help ⌘ WriteOut ⌘ Read File ⌘ Prev Pg ⌘ Cut Text ⌘ Cur Pos
⌘ Exit ⌘ Justify ⌘ Where is ⌘ Next Pg ⌘ UnCut Text ⌘ To Spell

4. C Program - Code (Continued)

```
UW PICO 5.09 File: day13.cpp

    if (digit % 2 == 0)
        even_dig++;
    else
        odd_dig++;

    temp = temp / 10;
}

printf("Even_dig = %d\n", even_dig);
printf("Odd_dig = %d\n", odd_dig);
break;

case 2:
    sum_even = 0;
    sum_odd = 0;
    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;

        if (digit % 2 == 0)
            sum_even = sum_even + digit;
        else
            sum_odd = sum_odd + digit;

        temp = temp / 10;
    }

    printf("Sum_even = %d\n", sum_even);
    printf("Sum_odd = %d\n", sum_odd);
    break;

default:
    printf("Invalid choice!\n");
}

return 0;
}
```

⌘ Get Help ⌘ WriteOut ⌘ Read File ⌘ Prev Pg ⌘ Cut Text ⌘ Cur Pos
⌘ Exit ⌘ Justify ⌘ Where is ⌘ Next Pg ⌘ UnCut Text ⌘ To Spell

5. Compilation, Execution and Testing

The program was compiled and executed for both menu choices. The outputs match the given examples and prepared test cases.

```
bhavya@Bhavyas-MacBook-Air ~ % nano day13.cpp
bhavya@Bhavyas-MacBook-Air ~ % clang day13.cpp -o day13
bhavya@Bhavyas-MacBook-Air ~ % ./day13
Enter a positive integer: 12845

MENU
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 1
Even_dig = 3
Odd_dig = 2
bhavya@Bhavyas-MacBook-Air ~ % ./day13
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 2
Sum_even = 6
Sum_odd = 9
bhavya@Bhavyas-MacBook-Air ~ %
```

6. Result

The menu-driven C program using switch-case was compiled and executed successfully. It correctly counts even and odd digits and finds the sums of even and odd digits separately.